Lab Activity: MBR, GPT and File System Analysis (DFIR)

1. Aim

To analyze disk partitioning schemes (**MBR and GPT**) and file systems (**NTFS and FAT**) to understand their structure, working principles, and forensic significance using **read-only analysis techniques**.

2. Objectives

- To identify the disk partitioning scheme (MBR or GPT).
- To analyze the structure of MBR and GPT.
- To understand NTFS and FAT file system internals.
- To observe forensic artifacts and deleted file indicators.
- To correlate partitioning and file system analysis in DFIR investigations.

3. Tools Required

- Windows Operating System
- PowerShell
- Disk Management Utility
- HxD Hex Editor (**Read-only mode**)
- FTK Imager (**Optional – Read-only**)

4. Safety Instructions

- Always open disks and partitions in **read-only mode**.
- Do not modify, overwrite, or repair disk structures.
- Perform analysis only on **personal systems**.
- Take screenshots of observations for documentation.
- This experiment is strictly for **educational purposes**.

5. Theory Overview

MBR (Master Boot Record)

- Located in the **first 512 bytes** of the disk.
- Contains:
  - Bootloader code
  - Partition table (4 entries)
  - Boot signature **55 AA**
- Supports up to **2 TB** disk size and **4 primary partitions**.
- Important in forensics to identify partition layout and boot behavior.

GPT (GUID Partition Table)

- Used with **UEFI firmware**.
- Contains:
  - Protective MBR
  - Primary GPT Header
  - Partition Entry Array
  - Backup GPT Header
- Supports **large disks** and up to **128 partitions**.
- Provides redundancy and better integrity for forensic analysis.

File Systems

FAT (File Allocation Table)

- Simple file system.
- Uses FAT table to track cluster allocation.
- Deleted files are often recoverable.
- Commonly found on USB drives and memory cards.

NTFS (New Technology File System)

- Advanced Windows file system.
- Uses **Master File Table (MFT)**.
- Supports journaling, permissions, large files.
- Rich source of forensic metadata.

6. Procedure

Task 1: Identify Disk Partitioning Scheme

Open PowerShell and execute:

```
Get-Disk
```

Record:

- Disk number
- Disk size
- Partition style (MBR / GPT)

Task 2: Analyze MBR / GPT Using HxD

1. Open HxD as Administrator.
2. Go to **Extras → Open Disk**.
3. Select **Physical Disk** and enable **Read-only mode**.

If Disk is MBR:

- Identify boot code region.
- Locate partition table.
- Verify MBR signature **55 AA**.

If Disk is GPT:

- Observe Protective MBR (partition type EE).
- Move to next sector.
- Verify GPT header signature **EFI PART**.

Task 3: Identify File System Type

Using PowerShell:

```
Get-Volume
```

Note:

- Drive letter
- File system type (NTFS / FAT)
- Volume size

Task 4: File System Analysis

For NTFS:

- Locate the NTFS boot sector.
- Identify:
  - Bytes per sector
  - Sectors per cluster
- Locate and observe **one MFT entry**.

For FAT:

- Locate the FAT boot sector.
- Identify:
  - FAT table
  - One directory entry
- Observe deleted file marker (**E5**).

Task 5: Deleted File Observation

- **NTFS:**
  Observe deleted entries in the MFT and note remaining metadata.
- **FAT:**
  Observe directory entries where the first byte is **E5**, indicating deletion.

7. Observation Table

Students should record:

- Disk number
- Disk size
- Partition style (MBR / GPT)
- File system type (NTFS / FAT)
- Boot sector observations
- Deleted file indicators

8. Important Questions (Total: 7)

1. Explain the structure and forensic importance of the **Master Boot Record (MBR)**.
2. What is the purpose of the **Protective MBR** in GPT disks?
3. Differentiate between **MBR and GPT** from a digital forensic perspective.
4. Explain how **Logical Block Addressing (LBA)** helps in partition analysis.
5. How do **SSD technologies such as TRIM and Garbage Collection** affect forensic investigations?
6. **(From FAT)** How does the FAT file system indicate a deleted file, and why is this important in digital forensics?
7. **(From NTFS)** What is the role of the **Master File Table (MFT)** in NTFS?

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